

IND320 - Reservoir data project

This report is a PDF backup of the Jupyter Notebook analysis. The corresponding notebook and Streamlit app are included in the same project repository.

GitHub repository: https://github.com/YOUR_USERNAME/ind320-reservoir-dashboard

Streamlit app: https://YOUR_PROJECT.streamlit.app/

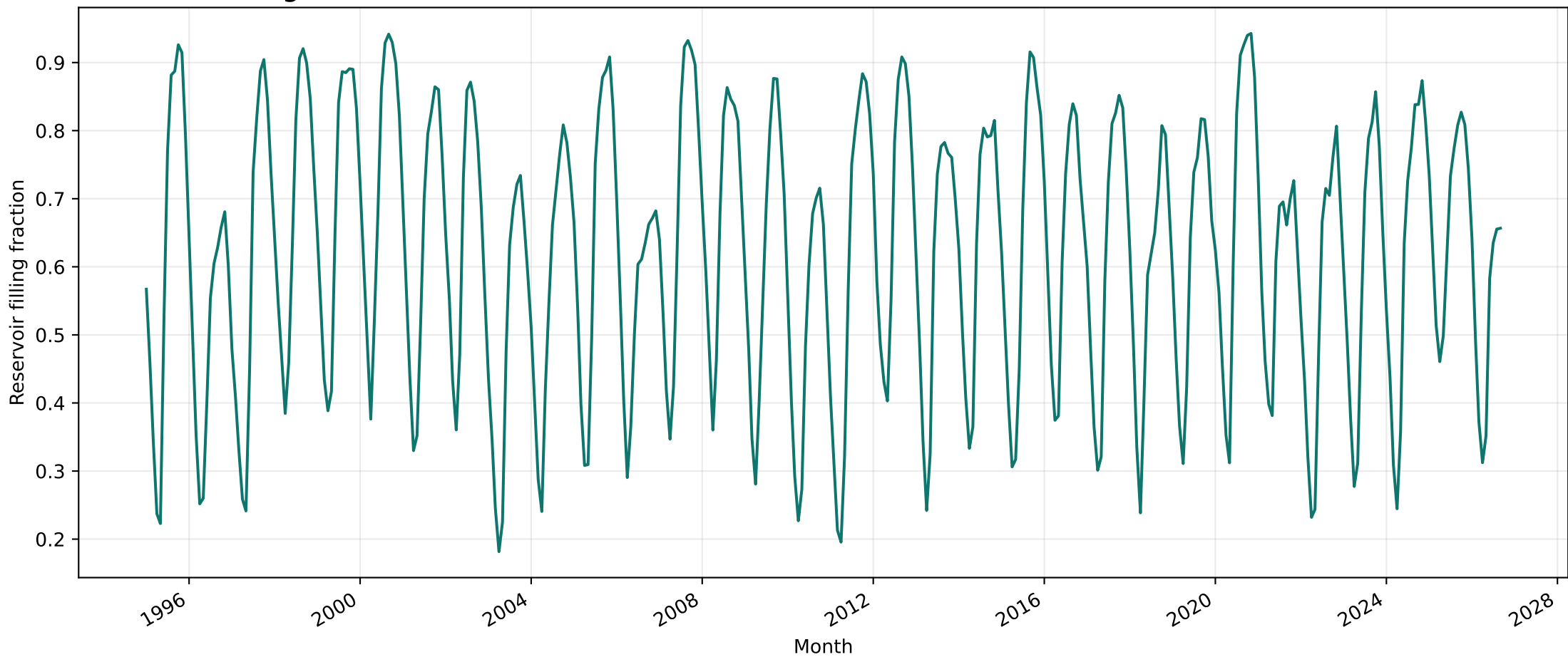
Loaded 14,877 rows and 11 columns. The date range is 1995-01-08 to 2026-09-06.

The data is weekly and is aggregated to monthly means for the time-series plots. The combined plot uses min-max scaling because the measurement columns have different units.

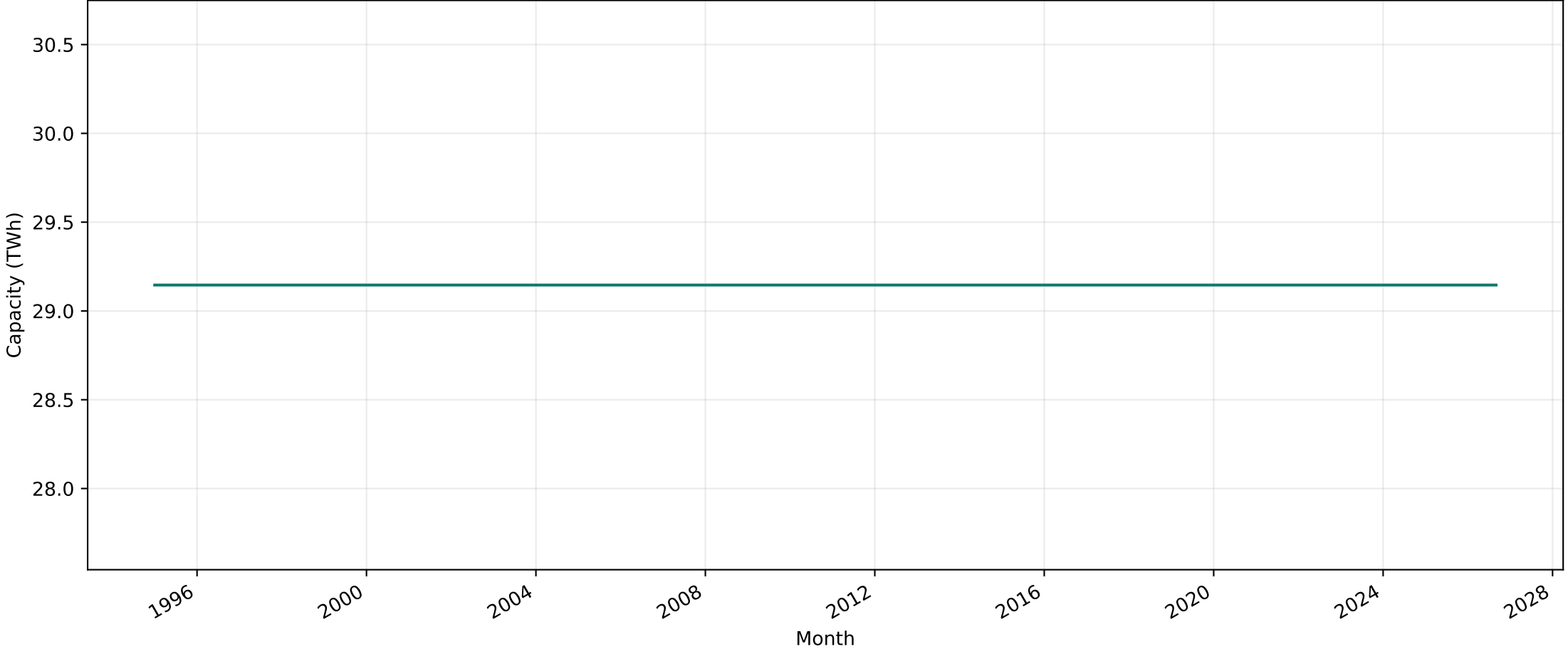
Renamed headers and data overview

English column name	Data type	Non-null values
date	datetime64[ns]	14877
area_type	object	14877
area_number	int64	14877
iso_year	int64	14877
iso_week	int64	14877
reservoir_filling_fraction	float64	14877
capacity_TWh	float64	14877
stored_energy_TWh	float64	14877
next_publication_date	datetime64[ns]	3555
filling_fraction_previous_week	float64	14877
change_in_filling_fraction	float64	14877

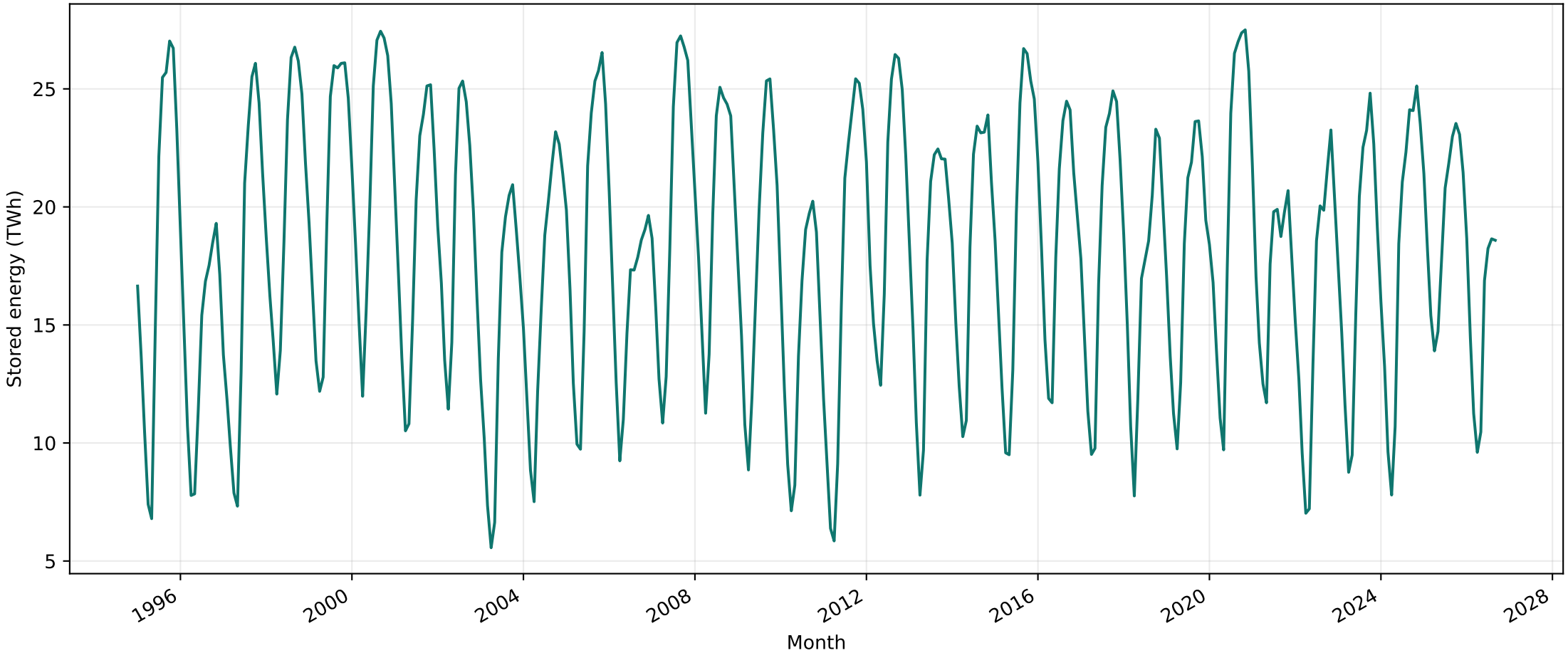
Reservoir filling fraction



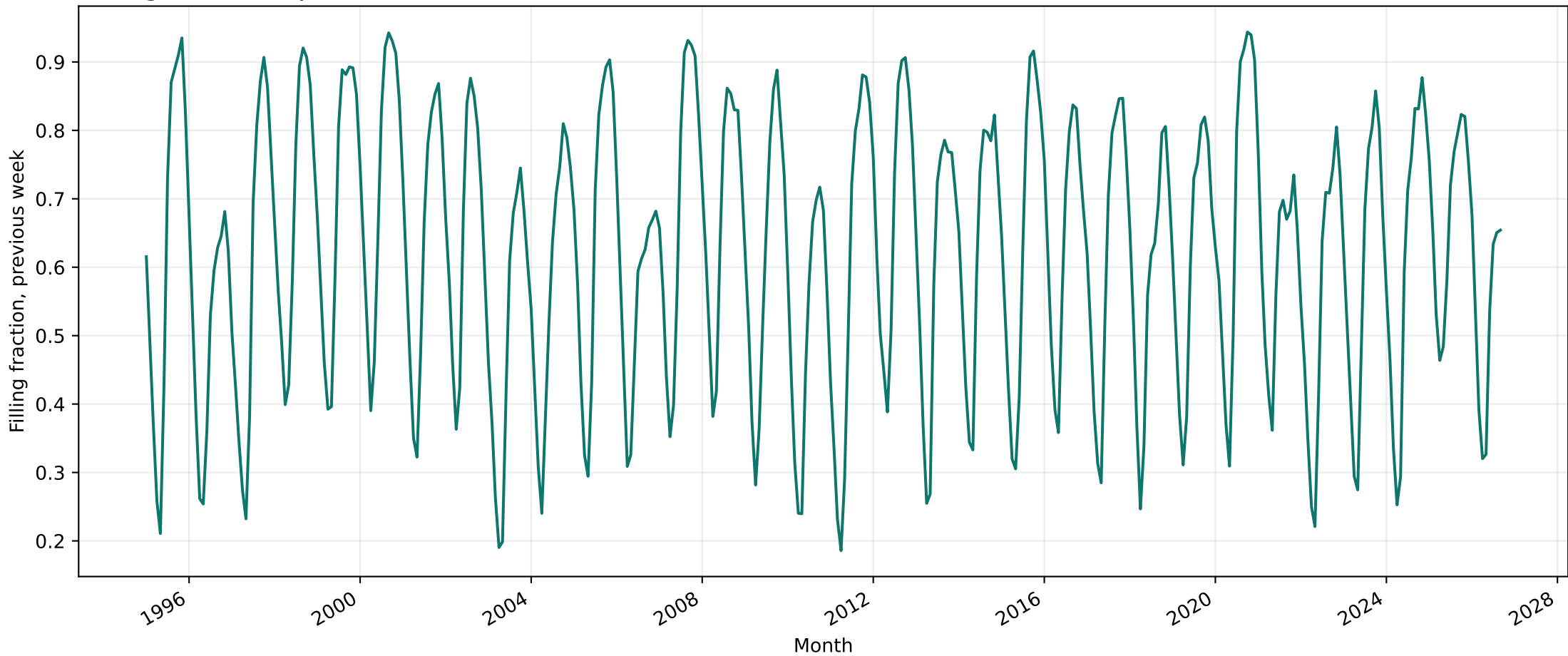
Capacity (TWh)



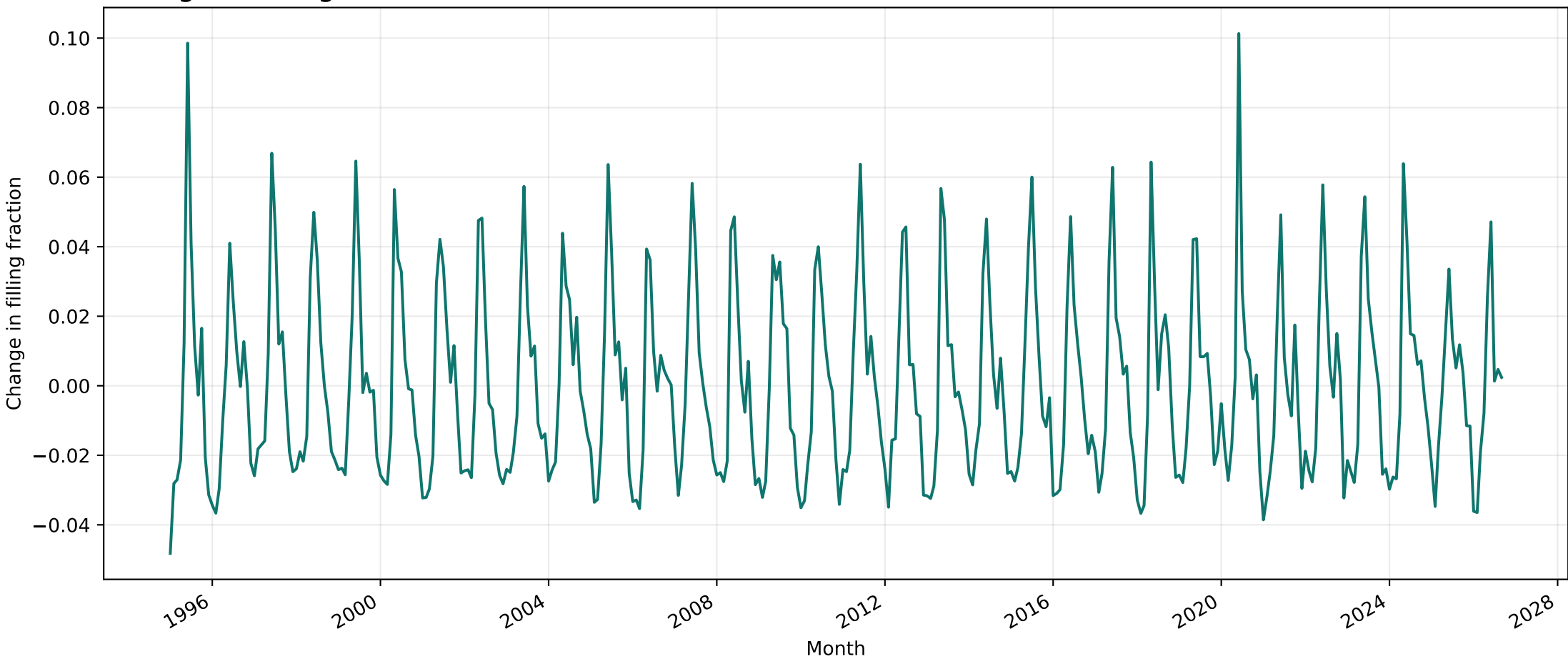
Stored energy (TWh)



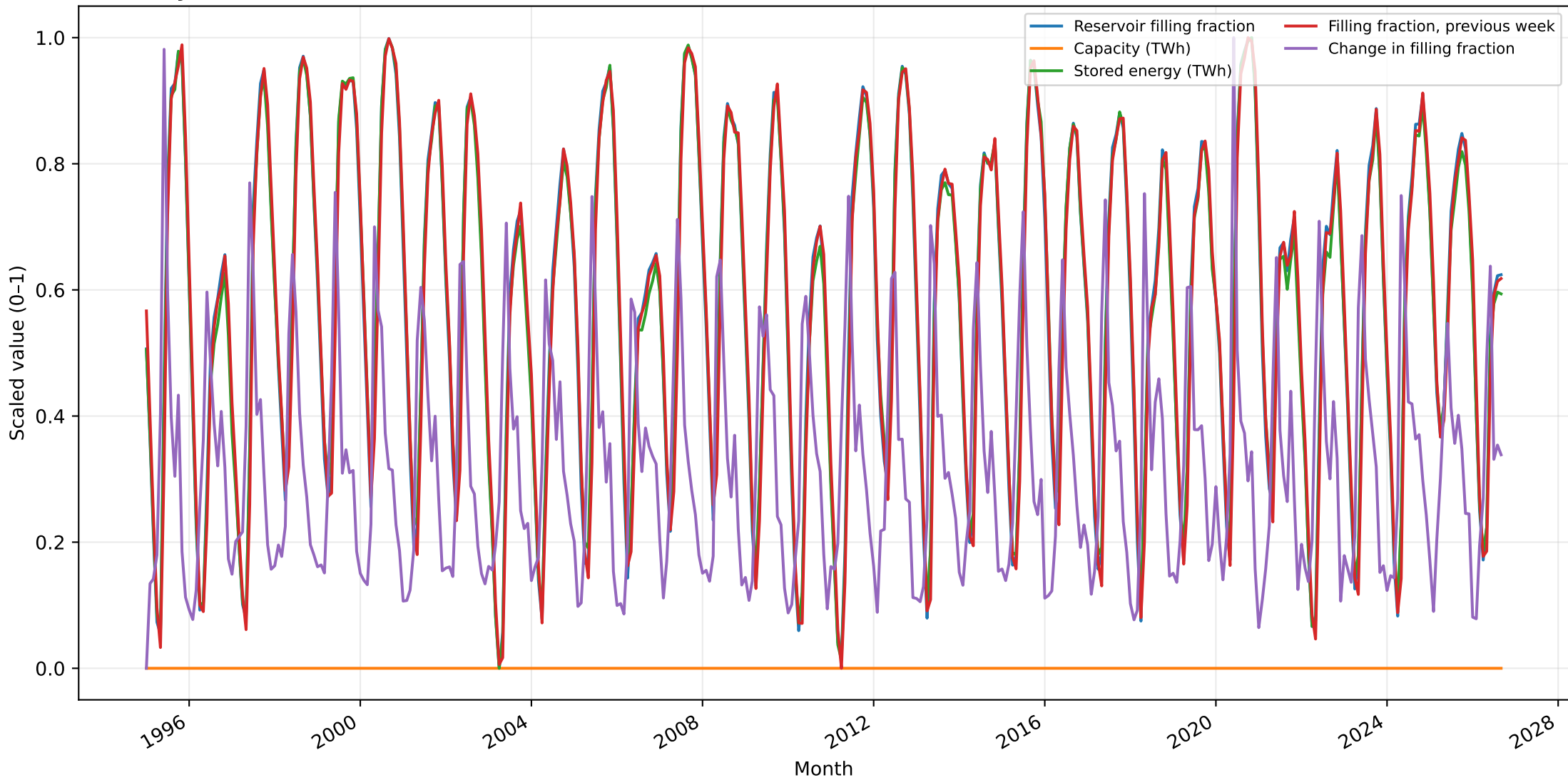
Filling fraction, previous week



Change in filling fraction



Monthly reservoir measurements, min-max scaled



AI usage and compulsory work log

AI usage: I used ChatGPT as a programming assistant to discuss the assignment requirements, suggest a clear project structure, explain Pandas and Streamlit syntax, and help check the code for reproducibility. I reviewed and adapted the suggestions, ran the code locally, and remain responsible for the final interpretation, plots and submission.

Work log: I started by reading `reservoirs.csv` with Pandas and checking its shape, data types and date range. I renamed the short Norwegian headers to understandable English names, parsed the date fields, and sorted the observations. Because the observations are weekly and contain several area groups, I calculated monthly means for five measurement variables. I plotted each measurement separately and then used min-max scaling for a combined plot.

I created a four-page Streamlit app with a home page, data table, plot page and About page. The data page contains one row per imported column and a `LineChartColumn` for the first month. The plot page has a `selectbox` and `select_slider`. The CSV read is cached for faster interaction. The notebook is for transparent analysis, while Streamlit presents the result interactively. Before submission, replace the placeholder URLs and personalise this log with any details from your own run.